battery is about to run out. The robot may then move towards an electrical socket, plug itself in and recharge its battery. However, throughout this process the robot doesn't experience anything. In contrast, a human being depleted of energy feels hunger and craves to stop this unpleasant sensation. That's why we say that humans are conscious beings and robots aren't, and why it is a crime to make people work until they collapse from hunger and exhaustion, whereas making robots work until their batteries run out carries no moral opprobrium.

And what about animals? Are they conscious? Do they have subjective experiences? Is it okay to force a horse to work until he collapses from exhaustion? As noted earlier, the life sciences currently argue that all mammals and birds, and at least some reptiles and fish, have sensations and emotions. However, the most up-to-date theories also maintain that sensations and emotions are biochemical data-processing algorithms. Since we know that robots and computers process data without having any subjective experiences, maybe it works the same with animals? Indeed, we know that even in humans many sensory and emotional brain circuits can process data and initiate actions completely unconsciously. So perhaps behind all the sensations and emotions we ascribe to animals – hunger, fear, love and loyalty – lurk only unconscious algorithms rather than subjective experiences?[2]

This theory was upheld by the father of modern philosophy, René Descartes. In the seventeenth century Descartes maintained that only humans feel and crave, whereas all other animals are mindless automata, akin to a robot or a vending machine. When a man kicks a dog, the dog experiences nothing. The dog flinches and howls automatically, just like a humming vending machine that makes a cup of coffee without feeling or wanting anything.

This theory was widely accepted in Descartes' day. Seventeenth-century doctors and scholars dissected live dogs and observed the working of their internal organs, without either anaesthetics or scruples. They didn't see anything wrong with that, just as we don't see anything wrong in opening the lid of a vending machine and observing its gears and conveyors. In the early twenty-first century there are still plenty of people who argue that animals